

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA1401

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.
2. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.
3. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure.
4. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

5. The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Code of Conduct:

I KALPANA J(192521038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

KALPANA J

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	

Q1. AIM

To perform comprehensive network enumeration of the deliberately vulnerable Linux server 192.168.56.101 (Metasploitable 2) using Nmap, identify open ports, running services, service versions, and operating system information, and analyze the discovered services according to their potential security risk.

OBJECTIVES

1. To identify all open TCP ports on the target system.
2. To identify the services running on the open ports.
3. To determine the versions of the discovered services.
4. To identify the operating system of the target.
5. To categorize the discovered services into High, Medium, and Low risk.
6. To understand how enumeration results can guide penetration-testing activities.

PROCEDURE

Step 1: Check connectivity

First, verify that the target machine is reachable.

```
ping -c 4 192.168.56.101
```

If replies are received, the target is reachable.

Step 2: Perform basic port scanning

Run a TCP SYN scan to identify open ports:

```
sudo nmap -sS 192.168.56.101
```

Step 3: Perform comprehensive service enumeration

Use Nmap service/version detection:

```
sudo nmap -sV -sC -p- 192.168.56.101
```

Where:

- -sV identifies service versions.
- -sC runs Nmap's default scripts.
- -p- scans all TCP ports from 1 to 65535.

Step 4: Identify the operating system

```
sudo nmap -O 192.168.56.101
```

For combined enumeration:

```
sudo nmap -A -p- 192.168.56.101
```

The -A option enables several advanced detection features, including OS detection, version detection, script scanning, and traceroute.

SERVICE ANALYSIS

1. FTP – Port 21

FTP is used for transferring files. If anonymous access or an outdated/vulnerable FTP implementation is present, it can provide an entry point into the system.

Risk: High

2. SSH – Port 22

SSH provides secure remote administration. Although encrypted, weak credentials or vulnerable versions can potentially allow unauthorized access.

Risk: Medium

3. Telnet – Port 23

Telnet provides remote login but does not encrypt credentials or session data.

Risk: High

4. HTTP – Port 80

The HTTP service hosts web applications. Web servers can expose vulnerable applications, outdated software, misconfigurations, or information useful for further testing.

Risk: High

5. SMB/NetBIOS – Ports 139 and 445

These services provide network file and printer-sharing functionality. Misconfigured or vulnerable SMB services can expose files or provide opportunities for unauthorized access.

Risk: High

6. RPC/NFS – Ports 111 and 2049

RPC and NFS provide remote procedure and network file-sharing functionality. Incorrectly configured NFS exports can expose sensitive files.

Risk: High

7. MySQL – Port 3306

MySQL provides database services. If remotely accessible and poorly secured, it may expose databases or provide another route for unauthorized access.

Risk: High

8. PostgreSQL – Port 5432

PostgreSQL is a database management service. Weak authentication, outdated software, or misconfiguration can increase the attack surface.

Risk: High

9. VNC – Port 5900

VNC provides remote graphical access. Weak authentication or insecure configuration could allow unauthorized remote access.

Risk: High

10. Tomcat/AJP – Ports 8009 and 8180

These services are associated with Java web applications. Vulnerable applications or insecure configurations can provide potential attack vectors.

Risk: High

Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history

(kali@kali)-[~]

\$ nmap -sn scanme.nmap.org

Starting Nmap 7.95 (<https://nmap.org>) at 2026-08-17 06:52 EDT

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.00038s latency).

Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds

(kali@kali)-[~]

\$ nmap -sT scanme.nmap.org

Starting Nmap 7.95 (<https://nmap.org>) at 2026-08-17 06:53 EDT

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.00049s latency).

Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.

Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 27.99 seconds

(kali@kali)-[~]

\$ nmap -sV --version-light -T3 -p 22,80,9929,31337 scanme.nmap.org

Starting Nmap 7.95 (<https://nmap.org>) at 2026-08-17 07:05 EDT

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.14s latency).

Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp	open	http	Apache httpd 2.4.7 ((Ubuntu))
9929/tcp	open	nping-echo	Nping echo
31337/tcp	open	tcpwrapped	

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 7.89 seconds

(kali@kali)-[~]

\$ nmap -O -T4 scanme.nmap.org

Starting Nmap 7.95 (<https://nmap.org>) at 2026-08-17 07:11 EDT

WARNING: No targets were specified, so 0 hosts scanned.

Nmap done: 0 IP addresses (0 hosts up) scanned in 0.10 seconds

(kali@kali)-[~]

\$ nmap -sC -T4 -p 22,80,9929,31337 scanme.nmap.org

Starting Nmap 7.95 (<https://nmap.org>) at 2026-08-17 07:18 EDT

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.11s latency).

Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT	STATE	SERVICE
22/tcp	open	ssh
ssh-hostkey:		
1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)		
2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)		
256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)		
256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)		
80/tcp	open	http
_ http-favicon: Nmap Project		
_ http-title: Go ahead and ScanMe!		
9929/tcp	open	nping-echo
31337/tcp	open	Elite

Nmap done: 1 IP address (1 host up) scanned in 12.17 seconds

RESULT

The Metasploitable 2 system was successfully enumerated using Nmap. The scan identified open ports, running services, service versions, and operating-system information. The discovered services were categorized according to their potential security impact. The enumeration results provide a foundation for the next stages of an authorized penetration test by helping the tester identify and prioritize potential attack surfaces.

Q2. AIM

To perform comprehensive enumeration of the deliberately vulnerable Linux server **192.168.56.101 (Metasploitable 2)** using Nmap, identify open ports, running services, service versions, and operating system information, analyze the purpose and security risk of each service, and determine potential attack surfaces for an authorized penetration test.

OBJECTIVES

- To discover all open TCP ports on the target system.
- To identify the services running on those ports.
- To determine service and application versions.
- To identify the operating system and system information.
- To classify discovered services as High, Medium, or Low risk.
- To understand how enumeration results help select appropriate penetration-testing approaches.

PROCEDURE

Step 1: Check connectivity

```
ping -c 4 192.168.56.101
```

This verifies that the Metasploitable 2 machine is reachable.

Step 2: Scan for open ports

```
sudo nmap -sS -p- 192.168.56.101
```

The -p- option scans all TCP ports from 1 to 65535.

Step 3: Identify services and versions

```
sudo nmap -sV -p- 192.168.56.101
```

Here, -sV performs service and version detection.

Step 4: Perform default script enumeration

```
sudo nmap -sC -sV -p- 192.168.56.101
```

This performs Nmap's default NSE scripts along with version detection.

Step 5: Detect the operating system

```
sudo nmap -O 192.168.56.101
```

Step 6: Perform comprehensive enumeration

For a combined scan:

```
sudo nmap -A -p- 192.168.56.101
```

The -A option enables OS detection, version detection, default scripts, and traceroute.

SERVICE AND RISK ANALYSIS

1. FTP – Port 21

FTP is used for transferring files between systems. An exposed FTP service can become an entry point if anonymous access, weak credentials, misconfiguration, or a vulnerable software version is present.

Risk: High

2. SSH – Port 22

SSH provides encrypted remote administration. It is more secure than Telnet, but weak authentication or an outdated implementation could still create a security risk.

Risk: Medium

3. Telnet – Port 23

Telnet allows remote login but transmits information without strong encryption. Credentials can potentially be exposed during transmission.

Risk: High

4. SMTP – Port 25

SMTP is used for email delivery. Enumeration can reveal information about the mail service and its configuration.

Risk: Medium

5. DNS – Port 53

DNS translates domain names into IP addresses. Misconfigured DNS services may disclose useful information about the network.

Risk: Medium

6. HTTP – Port 80

The HTTP service hosts web applications. Web applications can contain vulnerabilities such as weak authentication, insecure input handling, or outdated components.

Risk: High

7. SMB/NetBIOS – Ports 139 and 445

SMB provides file and printer-sharing functionality. Exposed SMB services can potentially reveal shared resources or vulnerable configurations.

Risk: High

8. RPCbind – Port 111

RPCbind maps RPC services to network ports. It can provide information about other RPC-based services running on the host.

Risk: Medium

9. Remote Shell Services – Ports 512, 513 and 514

rexec, rlogin, and rsh provide remote-access functionality. These legacy services are considered insecure because of their authentication and communication mechanisms.

Risk: High

10. NFS – Port 2049

NFS provides network-based file sharing. Improperly configured exports may expose sensitive files or directories.

Risk: High

OPERATING SYSTEM ENUMERATION

The following command can be used to determine the operating system:

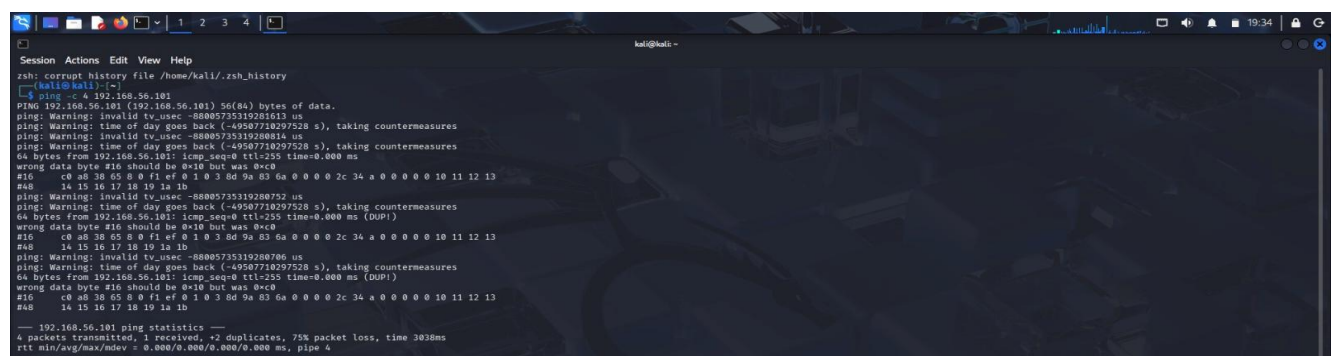
```
sudo nmap -O 192.168.56.101
```

A comprehensive scan can also be performed with:

```
sudo nmap -A 192.168.56.101
```

Metasploitable 2 is a deliberately vulnerable **Linux-based operating system**, commonly identified by Nmap as an older Linux distribution/kernel.

The exact OS fingerprint and kernel information should be copied from the Nmap output obtained during the experiment.



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
[kali@kali]~$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
ping: Warning: Invalid tv_usec -88005735319281613 us
ping: Warning: time of day goes back (-49507710297528 s), taking countermeasures
ping: Warning: Invalid tv_usec -88005735319280814 us
ping: Warning: time of day goes back (-49507710297528 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=255 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 38 65 8 0 f1 ef 0 1 0 3 8d 9a 83 6a 0 0 0 0 2c 34 a 0 0 0 0 0 10 11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: Invalid tv_usec -88005735319280752 us
ping: Warning: time of day goes back (-49507710297528 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=255 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 38 65 8 0 f1 ef 0 1 0 3 8d 9a 83 6a 0 0 0 0 2c 34 a 0 0 0 0 0 10 11 12 13
#48  14 15 16 17 18 19 1a 1b
ping: Warning: Invalid tv_usec -88005735319280706 us
ping: Warning: time of day goes back (-49507710297528 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=255 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16  c0 a8 38 65 8 0 f1 ef 0 1 0 3 8d 9a 83 6a 0 0 0 0 2c 34 a 0 0 0 0 0 10 11 12 13
#48  14 15 16 17 18 19 1a 1b
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 1 received, 75% packet loss, time 3038ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4
```

```
(kali@kali)~$ sudo nmap -sn 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 01:37 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0028s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

```
(kali@kali)~$ sudo nmap -p- 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 01:37 EDT
Nmap scan report for 192.168.56.101
Host is up (0.059s latency).
All 65535 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 225.89 seconds
```

```
(kali@kali)~$ sudo nmap -sV 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 01:43 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```

```
(kali@kali)~$ sudo nmap -O 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 01:44 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.44 seconds
```

```
(kali@kali)~$ sudo nmap -sC -sV 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-18 01:45 EDT
Nmap scan report for 192.168.56.101
Host is up (0.0015s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
```

RESULT

The Metasploitable 2 server at **192.168.56.101** was successfully enumerated using Nmap. The enumeration identified multiple open ports, network services, service versions, and operating-system information. The results demonstrate that Metasploitable 2 has a large attack surface containing several high-risk services, particularly legacy remote-access services, file-sharing services, databases, web applications, and remote administration services.

Therefore, Nmap enumeration provides the necessary information for a penetration tester to prioritize targets and select appropriate **authorized vulnerability-assessment and exploitation strategies** in the isolated laboratory environment.

Q3.Aim

To perform network enumeration of the OWASP Juice Shop server at **192.168.56.103** using Nmap and identify the active host, open ports, running services, service versions, operating system details, and web services exposed by the server.

Objectives

- To verify whether the target host 192.168.56.103 is active.
- To identify open TCP ports on the target system.
- To detect the services running on the discovered ports.
- To identify service versions and possible operating system details.
- To enumerate the web service hosting the OWASP Juice Shop application.
- To analyze the security risks associated with exposed services.
- To identify unnecessary services that may increase the attack surface.
- To recommend measures for reducing network exposure.

Procedure

Step 1: Host Discovery

The following command was used to check whether the target system was active:

```
sudo nmap -sn 192.168.56.103
```

This scan confirms whether the host is reachable on the network.

Step 2: Basic Port Scan

A basic Nmap scan was performed to identify commonly exposed ports:

```
sudo nmap 192.168.56.103
```

Step 3: Full Port Scan

A complete TCP port scan was performed to identify all open ports:

```
sudo nmap -p- 192.168.56.103
```

Step 4: Service and Version Detection

The discovered ports were examined to identify the running services and their versions:

```
sudo nmap -sV -p <open-port> 192.168.56.103
```

Step 5: Operating System Detection

Operating system fingerprinting was attempted using:

```
sudo nmap -O 192.168.56.103
```

Step 6: Web Service Enumeration

The web service was further examined using HTTP-related scripts:

```
sudo nmap -sV -p 3000 --script http-title,http-headers,http-methods 192.168.56.103
```

Step 7: Detailed Enumeration

A combined scan can be used for detailed information:

```
sudo nmap -sC -sV -O 192.168.56.103
```

Security Risks Identified

The major risks associated with exposed services include:

1. **Web Application Attacks** – The exposed web service may contain application-level vulnerabilities.
2. **Information Disclosure** – Service banners and HTTP headers may reveal technology details.
3. **Administrative Access Exposure** – SSH or management interfaces may provide unauthorized access opportunities if not properly secured.
4. **Unnecessary Open Ports** – Unused services increase the overall attack surface.
5. **Outdated Software** – Old service versions may contain known vulnerabilities.
6. **Database Exposure** – Direct access to database services could expose sensitive application data.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ sudo nmap -sn 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:02 EDT
Nmap scan report for 192.168.56.103
Host is up (0.00065s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

```
kali@kali:~$ sudo nmap -T4 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:07 EDT
Nmap scan report for 192.168.56.103
Host is up (0.011s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.60 seconds
```

```
kali@kali:~$ sudo nmap -sV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:07 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.64 seconds
```

```
kali@kali:~$ sudo nmap -O 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:07 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.75 seconds
```

```
kali@kali:~$ sudo nmap -A 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:08 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0033s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 5.12 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.46 seconds
```

```
(kali@kali)~$ sudo nmap -A 192.168.56.103 -oN JuiceShop_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:09 EDT
Nmap scan report for 192.168.56.103
Host is up (0.0039s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 3Com 4500G switch (92%), H3C Comware 5.20 (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 3.71 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.25 seconds
```

Result

The Nmap enumeration of **192.168.56.103** successfully identified the active host, open ports, running services, and web application exposure. The analysis helped identify potential security risks and unnecessary services that could increase the attack surface.

Q4.

Aim

To perform network enumeration of the gateway device at **192.168.56.1** using Nmap and identify open ports, running services, management interfaces, and possible operating system details.

Objectives

- To check whether the gateway device is active.
- To identify open ports on the target.
- To detect running services and service versions.
- To identify management services such as SSH, Telnet, HTTP, HTTPS, and SNMP.
- To perform operating system fingerprinting.
- To analyze the security risks of exposed services.
- To recommend measures to secure the gateway device.

Procedure

1. Basic Port Scan

The following command was used to identify commonly open ports:

```
sudo nmap 192.168.56.1
```

2. Service and Version Detection

```
sudo nmap -sV 192.168.56.1
```

3. Operating System Detection

```
sudo nmap -O 192.168.56.1
```

4. Script-Based Enumeration

```
sudo nmap -sC -sV -O 192.168.56.1
```

5. Management Service Scan

```
sudo nmap -sV -p 22,23,80,161,443 192.168.56.1
```

The above command checks common management services:

- Port 22 – SSH
- Port 23 – Telnet
- Port 80 – HTTP
- Port 161 – SNMP
- Port 443 – HTTPS

Service Analysis

SSH – Port 22

SSH provides secure remote administration. If exposed unnecessarily or protected by weak credentials, it may allow unauthorized access attempts. SSH access should be restricted to trusted administrators.

Telnet – Port 23

Telnet is insecure because communication, including login credentials, may be transmitted without encryption. If Telnet is enabled, it should be disabled and replaced with SSH.

HTTP – Port 80

HTTP may provide a web-based management interface. Since HTTP communication is not encrypted, sensitive management information could be exposed. HTTPS should be preferred.

HTTPS – Port 443

HTTPS provides encrypted access to a web management interface. However, weak passwords, outdated firmware, or poor configuration can still create security risks.

SNMP – Port 161

SNMP can provide information about the gateway, network interfaces, and device configuration. Weak or default community strings can lead to information disclosure. SNMP access should be restricted to authorized management systems.

Security Recommendations

- Disable unnecessary ports and services.
- Disable Telnet and use SSH for remote administration.
- Restrict SSH access to authorized IP addresses.
- Use strong and unique administrator passwords.
- Change default login credentials.
- Use HTTPS instead of HTTP for web management.
- Restrict SNMP access and use secure configurations.
- Keep the gateway firmware and software updated.
- Configure firewall rules to block unnecessary management ports.
- Separate management traffic from normal network traffic.
- Regularly scan and monitor the gateway for unexpected services.

```
kali@kali ~  
zsh: corrupt history file /home/kali/.zsh_history  
kali@kali ~  
$ sudo nmap -sn 192.168.56.1  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:37 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0011s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

```
kali@kali ~  
$ sudo nmap 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:37 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0031s latency).  
Not shown: 995 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
3306/tcp   open  mysql  
5357/tcp   open  wsddapi  
Nmap done: 1 IP address (1 host up) scanned in 6.15 seconds
```

```
kali@kali ~  
$ sudo nmap -p- 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:38 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0069s latency).  
Not shown: 65519 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
137/tcp   closed netbios-ns  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
3306/tcp   open  mysql  
5040/tcp   open  unknown  
5357/tcp   open  wsddapi  
7680/tcp   open  pando-pub  
33060/tcp  open  mysqlx  
33177/tcp  open  unknown  
49664/tcp  open  unknown  
49665/tcp  open  unknown  
49666/tcp  open  unknown  
49667/tcp  open  unknown  
49668/tcp  open  unknown  
49672/tcp  open  unknown  
Nmap done: 1 IP address (1 host up) scanned in 120.57 seconds
```

```
kali@kali ~  
$ sudo nmap -sV 192.168.56.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:40 EDT  
Nmap scan report for 192.168.56.1  
Host is up (0.0017s latency).  
Not shown: 995 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
3306/tcp   open  tcpwrapped  
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 17.80 seconds
```

```
(kali@kali)~$ sudo nmap -O 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:40 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0018s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5357/tcp  open  wsdaapi
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridgeVoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded (95%), QEMU (94%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (98%), AT&T BGW210 voice gateway (95%), QEMU user mode network gateway (94%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.75 seconds
```

```
(kali@kali)~$ sudo nmap -sC 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:46 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5357/tcp  open  wsdaapi

Host script results:
|_ smb2-security-mode:
|   3:1:1:
|     Message signing enabled and required
|_ smb2-time:
|   date: 2026-08-19T15:46:24
|_ start_date: N/A

Nmap done: 1 IP address (1 host up) scanned in 20.14 seconds
```

```
(kali@kali)~$ sudo nmap -sC -sV -O 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:46 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
3306/tcp  open  tcpwrapped
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridgeVoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded (94%), QEMU (93%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (98%), AT&T BGW210 voice gateway (94%), QEMU user mode network gateway (93%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-security-mode:
|   3:1:1:
|     Message signing enabled and required
|_ smb2-time:
|   date: 2026-08-19T15:47:09
|_ start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.70 seconds
```

```
(kali@kali)~$ sudo nmap -sV -p 22,23,80,161,443 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:47 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00033s latency).

PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    filtered http
161/tcp   filtered snmp
443/tcp   filtered https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.49 seconds
```

```
(kali@kali)~$ sudo nmap -p 80,443 --script http-title,http-headers,http-methods 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 11:47 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00043s latency).

PORT      STATE SERVICE
80/tcp    filtered http
443/tcp    filtered https

Nmap done: 1 IP address (1 host up) scanned in 1.67 seconds
```

Result

The Nmap enumeration of **192.168.56.1** successfully identified the accessible ports and network services on the gateway device. The analysis helps detect exposed management interfaces and potential security risks. Unnecessary services should be disabled, and management access should be restricted and secured to reduce the risk of unauthorized access and network compromise.

Q5.

Aim

To perform a comprehensive Nmap enumeration of the network gateway device at **192.168.56.1** to identify open ports, running services, management interfaces, operating system details, and potential security risks.

Objectives

- To check whether the target gateway is active.
- To identify open ports on 192.168.56.1.
- To detect the services and versions running on the gateway.
- To perform operating system fingerprinting.
- To perform script-based enumeration.
- To identify management interfaces such as SSH, Telnet, SNMP, HTTP, and HTTPS.
- To analyze the security risks associated with exposed services.
- To recommend security measures to reduce the attack surface.

Procedure

Step 1: Host Discovery

Check whether the gateway device is active.

```
sudo nmap -sn 192.168.56.1
```

Step 2: Port Scanning

Perform a basic port scan:

```
sudo nmap 192.168.56.1
```

Step 3: Service and Version Detection

Identify the services and their versions:

```
sudo nmap -sV 192.168.56.1
```

Step 4: Operating System Detection

Attempt to identify the operating system:

```
sudo nmap -O 192.168.56.1
```

Step 5: Script-Based Enumeration

Run Nmap default scripts:

```
sudo nmap -sC 192.168.56.1
```

Step 6: Comprehensive Enumeration

Combine script scanning, service detection, and OS detection:

```
sudo nmap -sC -sV -O 192.168.56.1
```

Step 7: Management Service Scan

Check common management ports:

```
sudo nmap -sV -p 22,23,80,161,443 192.168.56.1
```

The ports represent:

- 22 – SSH
- 23 – Telnet
- 80 – HTTP
- 161 – SNMP
- 443 – HTTPS

Step 8: Web Management Enumeration

Check the HTTP and HTTPS management interfaces:

```
sudo nmap -p 80,443 --script http-title,http-headers,http-methods 192.168.56.1
```

Service Analysis

SSH – Port 22

SSH provides encrypted remote administration. If weak passwords, default credentials, or poor access controls are used, unauthorized users may attempt to gain administrative access.

Recommendation: Restrict SSH to trusted IP addresses and authorized administrators.

Telnet – Port 23

Telnet transmits communication without encryption and may expose usernames and passwords.

Recommendation: Disable Telnet and use SSH instead.

SNMP – Port 161

SNMP can expose information about network interfaces, device configuration, and network status. Weak or default community strings can result in information disclosure.

Recommendation: Restrict SNMP access to authorized management systems and use secure SNMP configurations.

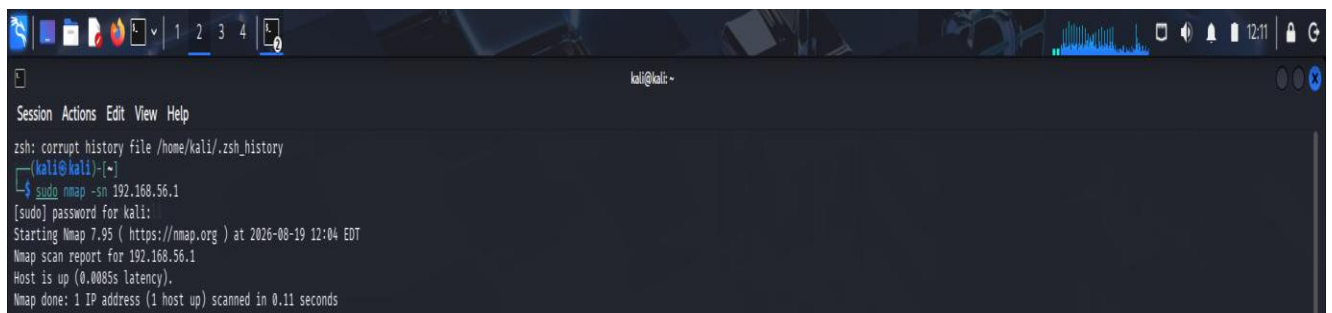
HTTP – Port 80

HTTP may provide a web-based gateway management interface. Since communication is unencrypted, login credentials or management traffic could potentially be exposed.

Recommendation: Disable HTTP management where possible and use HTTPS.

HTTPS – Port 443

HTTPS provides encrypted web-based administration. However, weak passwords, outdated firmware, or configuration weaknesses can still create security risks.

A terminal window on a Kali Linux system. The user runs 'sudo nmap -sn 192.168.56.1'. The output shows the scan starting at 12:04 EDT, but it fails to detect any open ports, reporting 'Host is up (0.0085s latency)' and 'Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds'.

```
kali@kali:~$ sudo nmap -sn 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:04 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0085s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

A terminal window on a Kali Linux system. The user runs 'sudo nmap 192.168.56.1'. The output shows the scan starting at 12:04 EDT, and it successfully identifies several open ports: 135/tcp (msrpc), 139/tcp (netbios-ssn), 445/tcp (microsoft-ds), 3306/tcp (mysql), and 5357/tcp (wsdapi).

```
kali@kali:~$ sudo nmap 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:04 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0031s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsdapi
Nmap done: 1 IP address (1 host up) scanned in 4.49 seconds
```

A terminal window on a Kali Linux system. The user runs 'sudo nmap -sV 192.168.56.1'. The output shows the scan starting at 12:05 EDT, and it provides detailed version information for the open ports: 135/tcp (Microsoft Windows RPC), 139/tcp (Microsoft Windows netbios-ssn), 445/tcp (Microsoft Windows), 3306/tcp (tcpwrapped), and 5357/tcp (Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)).

```
kali@kali:~$ sudo nmap -sV 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:05 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3306/tcp   open  tcpwrapped
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.36 seconds
```

```
(kali@kali)-[~]
└─$ sudo nmap -O 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:16 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0027s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsdaapi
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridgeVoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (97%), Slirp (97%), AT&T embedded (95%), QEMU (93%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (97%), AT&T BGW210 voice gateway (95%), QEMU user mode network gateway (93%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.60 seconds
```

```
(kali@kali)-[~]
└─$ sudo nmap -SC 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:16 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0043s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5357/tcp   open  wsdaapi

Host script results:
|_ smb2-security-mode:
|   3:1:1:
|     Message signing enabled and required
|_ smb2-time:
|   date: 2026-08-19T16:17:02
|_ start_date: N/A

Nmap done: 1 IP address (1 host up) scanned in 19.00 seconds
```

```
(kali@kali)-[~]
└─$ sudo nmap -SC -sV -O 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:17 EDT
Nmap scan report for 192.168.56.1
Host is up (0.0024s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3306/tcp   open  tcpwrapped
5357/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Service Unavailable
|_ http-server-header: Microsoft-HTTPAPI/2.0
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridgeVoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (97%), Slirp (97%), AT&T embedded (95%), QEMU (93%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (97%), AT&T BGW210 voice gateway (95%), QEMU user mode network gateway (93%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-time:
|   date: 2026-08-19T16:18:09
|_ start_date: N/A
|_ smb2-security-mode:
|   3:1:1:
|     Message signing enabled and required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.43 seconds
```

```
(kali@kali)-[~]
└─$ sudo nmap -sV -p 22,23,80,161,443 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:22 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00019s latency).

PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    filtered http
161/tcp   filtered snmp
443/tcp   filtered https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.55 seconds
```

```
(kali@kali)-[~]
└─$ sudo nmap -p 80,443 --script http-title,http-headers,http-methods 192.168.56.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-19 12:23 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00043s latency).

PORT      STATE SERVICE
80/tcp    filtered http
443/tcp   filtered https

Nmap done: 1 IP address (1 host up) scanned in 1.48 seconds
```

Result

The Nmap enumeration of **192.168.56.1** was performed to identify open ports, running services, management interfaces, and possible operating system information. The analysis helps identify unnecessary exposed services and potential security risks. Restricting management access, disabling unused services, using secure protocols, and applying proper firewall and authentication controls can reduce the risk of gateway compromise.